Agrigine

Cultivating the Future of Agriculture

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Problem Statement & Ideal User

PROBLEM STATEMENT

Farm data is scattered across a dozen sources, and the tools that predict from it are black boxes.

IDEAL USER

- Farmers
- Garderners
- Environmental Researchers
- Students
- Educators





Why Our Tech

DAILY & GENERALIZED DATA

County-level data for every day, 2000 to 2028, not just season totals.

AI COMPATIBILITY

Ask any AI assistant; it reads our data and cites the source.

CUSTOM DATA POINTS AND VISUALIZATIONS

Pick from 63 layers, build charts and reports, export anything.



How it Works

1

HOSTING

Static files on Cloudflare Pages, with the MCP server as a Worker. No app server, so it costs nothing to run.

2

BACKEND

A Python pipeline pulls eleven sources into compact files. Supabase Postgres holds accounts, with row-level security.

3

MACHINE LEARNING

Three scikit-learn notebooks test every model on ten held-out years, then pre-run all predictions into files the browser reads.

4

FRONTEND

Next.js and React with a deck.gl WebGL globe over NASA satellite imagery, and hand-written SVG charts.

Example Case #1

Sarah asks Claude, “which Iowa counties lost the most corn in 2012?” Claude calls Agrigine and answers with real numbers: Appanoose, 44.5 bushels per acre, 68.7% below its trend, 13 weeks in severe drought. She follows up, “pull their soil water storage and weekly drought,” gets a table with sources, and opens the map in Agrigine to see the pattern. No downloads, no joins, no black box.

LEARN MORE



THIS IS SARAH

A university soil scientist studying why some counties lose less corn in drought years. She connects Claude to Agrigine’s MCP server in one command.

Example Case #2

Dave uploads their spreadsheet to Agrigine, and it builds a model of their farm from their own yields, checked season by season against simply using their average. They ask Claude, “what should I plan for corn in 2027?” and get their farm’s predicted yield with a range, plus the safe planting date from 26 years of local freeze dates. Their records stay private, and they can delete them any time.



THIS IS SAM

A corn and soybean grower in Story County, Iowa, with three fields and 18 seasons of yield records in a spreadsheet.



Let’s Grow Together!

https://agrigine.pages.dev